ECON 5345 Homework 3 Report

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Question 1

a.

Question 2

a.

Question 3

I assume $e_t \sim \mathcal{N}(0,1)$ and $u_t \sim \mathcal{U}(-\sqrt{3}, \sqrt{3})$ for data generation. I checked the sample moment $\mathbb{E}[e_t u_t] = 0.0025$, consistent with orthogonality assumption.

a. The following results are for HP-filter.

Statistic	Value
Min	-0.5874
1st Qu.	-0.1134
Median	0.0092
Mean (μ^{HP})	0.0075
3rd Qu.	0.1215
Max	0.5501
SD (σ^{HP})	0.1707
$\mathbb{P}(\rho_{xy} > \mu^{HP} + 1\sigma^{HP})$	0.1550
$\mathbb{P}(\rho_{xy} > \mu^{HP} + 2\sigma^{HP})$	0.0240
$\mathbb{P}(\rho_{xy} > \mu^{HP} + 3\sigma^{HP})$	0.0005

Table 1: Summary statistics and upper-tail shares of simulated correlations for HP filter.

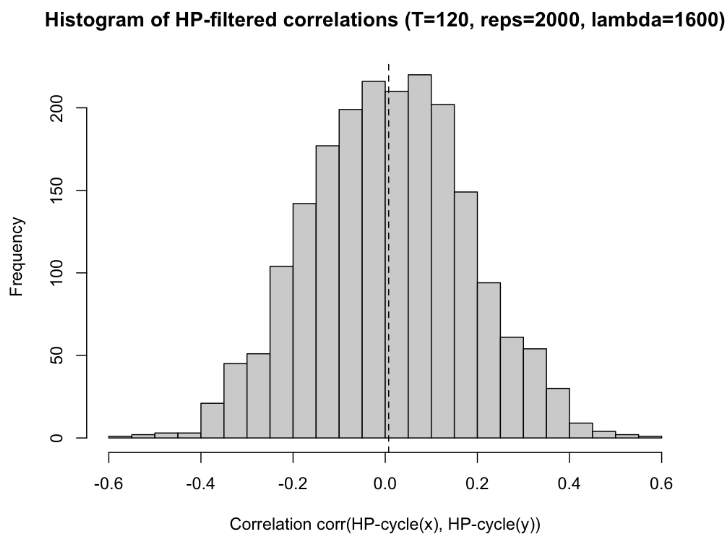

Figure 1: Histogram of HP-filtered correlations.

b. The following results are for Baxter-King filter.

Statistic	Value
Min	-0.6671
1st Qu.	-0.1432
Median	0.0135
Mean (μ^{BK})	0.0092
3rd Qu.	0.1619
Max	0.7416
SD (σ^{BK})	0.2171
$\mathbb{P}(\rho_{xy} > \mu^{BK} + 1\sigma^{BK})$	0.1655
$\mathbb{P}(\rho_{xy} > \mu^{BK} + 2\sigma^{BK})$	0.0225
$\mathbb{P}(\rho_{xy} > \mu^{BK} + 3\sigma^{BK})$	0.0005

Table 2: Summary statistics and upper-tail shares of BK-filtered correlations (T=120, reps=2000, K=12, periods [6,32]).

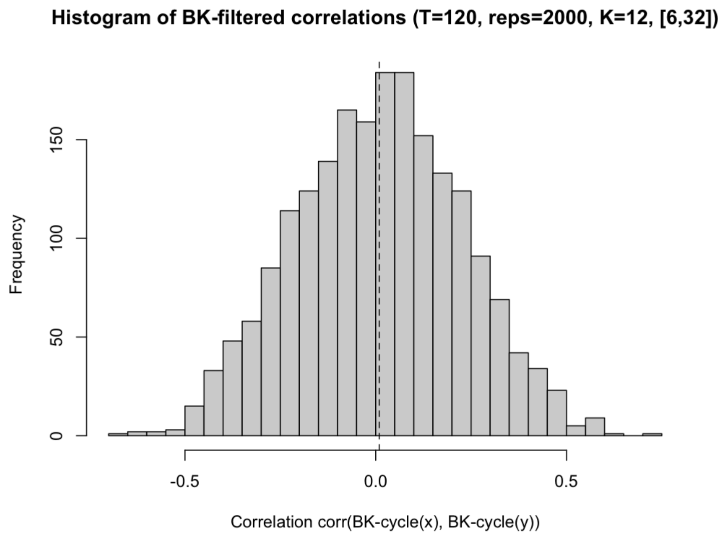

Figure 2: Histogram of BK-filtered correlations.

c. Discussion: In both cases, the mean correlation is close to zero, as expected under independence. However, the distribution of ρ_{xy} is dispersed indicated by the standard deviation values 0.1707 (HP) and 0.2171 (BK), and similar to a normal distribution, indicated by $\mathbb{P}(\rho_{xy} > \mu + z\sigma)$ values. For extreme values, correlations as large as 0.55 (HP) and 0.74 (BK) occur in the simulations.

Question 4

a. The gain of H^+ is plotted in 3. The highest value is at around $|\omega| = 0.25 \in \left(\frac{2\pi}{32}, \frac{2\pi}{6}\right)$ which is in the range of the business cycle frequency. Therefore, $\{c_t\}$ will have more variation at the business cycle frequencies.

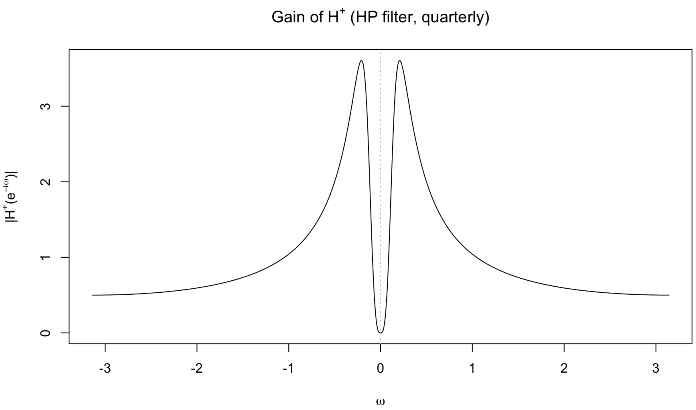

Figure 3: Gain of H^+

b. Since $c_t = H^+(L)y_t$, we have

$$s_c(\omega) = \left|H^+\left(e^-i\omega\right)\right|^2 s_y(\omega).$$

Since $y_t = x_t - x_{t-1} = e_t \sim WN(0, \sigma^2)$, we have $s_y(\omega) = \frac{\sigma^2}{2\pi}$. Therefore, the spectrum of $\{c_t\}$ is

$$s_c(\omega) = \frac{\sigma^2}{2\pi}\left|H^+\left(e^-i\omega\right)\right|^2$$

c. By definition, $s_c(\omega) = \frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)e^{-i\omega j}$. Multiply both sides by $e^{i\omega k}$ for some integer k and integrate over $(-\pi, \pi]$:

$$\begin{aligned}
\int_{-\pi}^{\pi} s_c(\omega)e^{i\omega k}d\omega &= \int_{-\pi}^{\pi}\frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)e^{i\omega(k-j)}d\omega \\
&= \frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j)\int_{-\pi}^{\pi}e^{i\omega(k-j)}d\omega \qquad \text{by LDCT} \\
&= \frac{1}{2\pi}\sum_{j=-\infty}^{+\infty}\gamma_c(j) \times 2\int_{0}^{\pi}\cos\left((k-j)\omega\right)d\omega \\
&= \gamma_c(k),
\end{aligned}$$

where the last line is by the periodicity of the $\cos(\cdot)$ function.

d. The table and figure below show the results of autocovariances.

k	$\gamma_c(k)$
0	1.668259408
1	1.203349899
2	0.807403465
3	0.477351924
4	0.208784655
5	-0.003618249
6	-0.165723093
7	-0.283644620
8	-0.363507014
9	-0.411257601
10	-0.432526914

Table 3: Autocovariances of $\{c_t\}$.

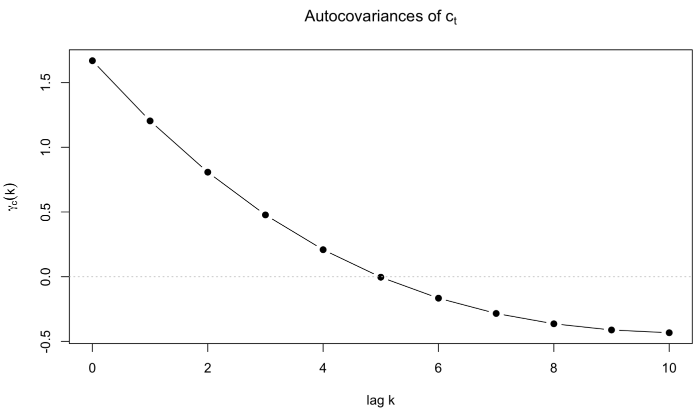

Figure 4: Autocovariances of c_t.